NIOS Secondary (Class 10) Science & Technology (212)

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

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High-Probability Predicted Mock Examination (85 Marks)

Time: 2½ Hours | Maximum Marks: 85

> **Instructions for the Student**:
> Attempt this paper on **Day 7** in exam-like conditions.
> Time yourself for 2.5 hours. Check your answers against the complete step-by-step solutions at the end.

SECTION A (Objective Type Questions — 42 Marks)

Q1 to Q17: Multiple Choice Questions (1 Mark Each)

1. The SI unit of electric resistance is:
   (A) Volt
   (B) Ampere
   (C) Ohm
   (D) Joule

2. The power of a convex lens of focal length $50\ \text{cm}$ is:
   (A) $+2.0\ \text{D}$
   (B) $-2.0\ \text{D}$
   (C) $+0.5\ \text{D}$
   (D) $+5.0\ \text{D}$

3. A body of mass $5\ \text{kg}$ is moving with an acceleration of $4\ \text{m/s}^2$. The net force acting on it is:
   (A) $1.25\ \text{N}$

(B) $9\ \mathrm{N}$
(C) $20\ \mathrm{N}$
(D) $0.8\ \mathrm{N}$

4. The acceleration due to gravity on the Moon is:
   (A) Equal to that on Earth
   (B) $\frac{1}{6}^{\text{th}}$ of that on Earth
   (C) 6 times that on Earth
   (D) Zero

5. Which defect of vision is corrected by using a concave lens?
   (A) Hypermetropia
   (B) Myopia
   (C) Presbyopia
   (D) Astigmatism

6. Three resistors of $2\ \Omega$ each are connected in series. Their equivalent resistance is:
   (A) $\frac{2}{3}\ \Omega$
   (B) $2\ \Omega$
   (C) $4\ \Omega$
   (D) $6\ \Omega$

7. The chemical formula of Bleaching Powder is:
   (A) CaO
   (B) $\mathrm{CaOCl_2}$
   (C) $\mathrm{Ca(OH)_2}$
   (D) $\mathrm{CaCl_2}$

8. Which of the following substances has a pH greater than 7?
   (A) Lemon juice
   (B) Gastric juice
   (C) Milk of Magnesia
   (D) Pure water

9. Which metal is stored under kerosene oil to prevent spontaneous reaction with air?
   (A) Iron
   (B) Magnesium
   (C) Sodium
   (D) Copper

10. The non-metal which is a good conductor of electricity is:
    (A) Diamond
    (B) Graphite
    (C) Sulphur
    (D) Phosphorus

11. The general molecular formula for Alkanes is:
    (A) $\mathrm{C}_n\mathrm{H}_{2n}$
    (B) $\mathrm{C}_n\mathrm{H}_{2n+2}$
    (C) $\mathrm{C}_n\mathrm{H}_{2n-2}$
    (D) $\mathrm{C}_n\mathrm{H}_{2n+1}\mathrm{OH}$

12. The functional group present in ethanoic acid ($\mathrm{CH_3COOH}$) is:
    (A) $-\mathrm{OH}$
    (B) $-\mathrm{CHO}$
    (C) $-\mathrm{COOH}$
    (D) $>\mathrm{C}=\mathrm{O}$

13. The structural and functional filtration unit of the human kidney is:
    (A) Neuron
    (B) Nephron
    (C) Alveolus
    (D) Villi

14. During anaerobic respiration in yeast, glucose is converted into:
    (A) Lactic acid + Energy
    (B) Ethanol + $\mathrm{CO_2}$ + Energy
    (C) $\mathrm{CO_2}$ + $\mathrm{H_2O}$ + Energy
    (D) Pyruvate only

15. The phenotypic ratio obtained in the F_2 generation of Mendel's monohybrid cross is:
    (A) $1:2:1$
    (B) $3:1$
    (C) $9:3:3:1$
    (D) $2:1$

16. Which part of the human eye controls the amount of light entering the eye?
    (A) Cornea
    (B) Iris

(C) Retina
(D) Ciliary muscles

17. The primary chemical responsible for stratospheric ozone depletion is:
(A) Carbon dioxide
(B) Methane
(C) Chlorofluorocarbon (CFC)
(D) Sulphur dioxide

Q18 to Q27: Objective Questions (2 Marks Each — 1 Mark Per Sub-Part)

1. **Fill in the blanks**:
(i) The device used to measure electric current in a circuit is called `_____`.
(ii) Work done is zero when the angle between force and displacement is `_____` degrees.

2. **True or False**:
(i) Mass of an object changes from place to place. (`TRUE / FALSE`)
(ii) Action and reaction forces act on the same body. (`TRUE / FALSE`)

3. **Fill in the blanks**:
(i) An electric fuse wire has high resistance and a `_____` melting point.
(ii) The commercial unit of electrical energy ($1\ \mathrm{kWh}$) is equal to `_____` Joules.

4. **True or False**:
(i) A concave mirror always forms an erect and virtual image. (`TRUE / FALSE`)
(ii) Violet light suffers maximum deviation when passed through a glass prism. (`TRUE / FALSE`)

5. **Fill in the blanks**:
(i) The chemical formula of Baking Soda is `_____`.
(ii) Plaster of Paris is obtained by heating Gypsum at a temperature of `_____` K.

6. **True or False**:
(i) Pure gold is 24 carats and is too soft for making jewelry. (`TRUE / FALSE`)
(ii) Roasting is the process of heating carbonate ores in limited air. (`TRUE / FALSE`)

7. **Fill in the blanks**:
(i) The unique ability of carbon atoms to form long chains with other carbon atoms is called `_____`.
(ii) Saponification is the chemical reaction used to prepare `_____`.

8. **True or False**:
   (i) Deoxygenated blood from the body tissues enters the left atrium of the human heart. (`TRUE / FALSE`)
   (ii) Insulin is secreted by the thyroid gland. (`TRUE / FALSE`)

9. **Fill in the blanks**:
   (i) In human beings, the normal sex chromosome complement of a female is `_____` .
   (ii) Fertilization of the human egg occurs in the `_____` .

10. **Match the following**:
Column I (Indicator / Phenomenon)	Column II (Observation / Effect)
(i) Phenolphthalein in base	(a) Pink colour
(ii) DDT accumulation in food chain	(b) Biomagnification

Q28: Case Study Question (5 Marks — 1 Mark Per Sub-Part)

1. **Case Study**: A student sets up an electric circuit consisting of an electric heater rated $1000\ \mathrm{W}$, an electric bulb rated $100\ \mathrm{W}$, and a television rated $100\ \mathrm{W}$ operating on a $220\ \mathrm{V}$ domestic power line.
   (i) In which combination (series or parallel) are domestic household appliances connected?
   (ii) Calculate the total electric power consumed when all three appliances are switched on together.
   (iii) What is the total electric current drawn from the $220\ \mathrm{V}$ line?
   (iv) If these appliances run for 5 hours daily, calculate the electrical energy consumed in kWh (units) per day.
   (v) Name the safety wire in domestic circuits that protects users from electric shocks.

SECTION B (Subjective Questions — 43 Marks)

Q29 to Q37: Very Short Answer Questions (2 Marks Each — 18 Marks)

1. State Newton's Second Law of Motion and write its mathematical formula.
2. State the difference between Mass and Weight (any two differences).
3. An object is placed at a distance of $30\ \mathrm{cm}$ in front of a concave mirror of focal length $15\ \mathrm{cm}$. Find the image distance (v).

4. Define 1 Dioptre of power of a lens. What type of lens has a power of $-1.5\ \mathrm{D}$?
5. Write the balanced chemical equation for the preparation of Bleaching Powder from slaked lime. State one of its uses.
6. Why is sodium metal kept immersed in kerosene oil?
7. What is an homologous series? Give the names and molecular formulas of the first two members of the alkane series.
8. Draw a neat labeled diagram of a nerve cell (neuron).
9. State the 10% law of energy transfer in an ecosystem.

Q38 to Q42: Short Answer Questions (3 Marks Each — 15 Marks)

1. Describe the three pathways of glucose breakdown during cellular respiration in:
   (i) Yeast (anaerobic)
   (ii) Human muscle cells (lack of oxygen)
   (iii) Mitochondria (aerobic)
2. Write the chemical formula, preparation equation, and two important uses of Plaster of Paris (POP).
3. Explain Mendel's monohybrid cross between a pure tall pea plant (TT) and a pure dwarf pea plant (tt). Show the Punnett square for the F_2 generation and state its phenotypic ratio.
4. (a) Define electric power.
   (b) An electric iron of resistance $20\ \Omega$ draws a current of $5\ \mathrm{A}$. Calculate the heat developed in 30 seconds.
5. Differentiate between Roasting and Calcination with one balanced chemical equation for each.

Q43 to Q44: Long Answer Questions (5 Marks Each — 10 Marks)

1. (a) What is Myopia? State its two causes.
   (b) Draw neat ray diagrams showing:
   (i) a myopic eye, and
   (ii) correction of myopia using an appropriate lens.
   (OR)
   (a) State Ohm's law. Draw a schematic circuit diagram for its verification in the laboratory.
   (b) Three resistors of resistances $3\ \Omega, 6\ \Omega$, and $2\ \Omega$ are given. How would you connect them to obtain an equivalent resistance of $4\ \Omega$? Show the calculation.

2. (a) Describe the structure of a nephron with a labeled diagram.
   (b) Briefly explain the three main steps of urine formation in the human kidney.

(OR)

(a) Explain why carbon forms a vast number of compounds (explain catenation and tetravalency).

(b) Explain the cleansing action of soaps with the help of a labeled diagram of a micelle.

COMPLETE STEP-BY-STEP SOLUTIONS & MARKING SCHEME

Section A Solutions

- **Q1**: (C) Ohm (Ω)
- **Q2**: (A) $+2.0\ \text{D}$ [$P = \frac{1}{0.5\ \text{m}} = +2.0\ \text{D}$]
- **Q3**: (C) $20\ \text{N}$ [$F = ma = 5 \times 4 = 20\ \text{N}$]
- **Q4**: (B) $\frac{1}{6}^{\text{th}}$ of that on Earth
- **Q5**: (B) Myopia
- **Q6**: (D) $6\ \Omega$ [$R_s = 2 + 2 + 2 = 6\ \Omega$]
- **Q7**: (B) CaOCl_2
- **Q8**: (C) Milk of Magnesia [pH ≈ 10]
- **Q9**: (C) Plaster of Paris
- **Q10**: (C) Strongly acidic
- **Q11**: (B) $\text{C}_n\text{H}_{2n+2}$
- **Q12**: (C) $-\text{COOH}$ (Carboxylic acid)
- **Q13**: (B) Nephron
- **Q14**: (B) Ethanol + CO_2 + Energy
- **Q15**: (B) $3:1$
- **Q16**: (B) Iris
- **Q17**: (C) Chlorofluorocarbon (CFC)
- **Q18**: (i) ammeter | (ii) 90° [$W = Fs\cos 90^\circ = 0$]
- **Q19**: (i) FALSE [Mass is constant] | (ii) FALSE [Act on different bodies]
- **Q20**: (i) low | (ii) $3.6 \times 10^6\ \text{J}$
- **Q21**: (i) FALSE [Convex mirror is always virtual/erect] | (ii) TRUE
- **Q22**: (i) NaHCO_3 | (ii) $373\ \text{K}$ (100°C)
- **Q23**: (i) TRUE | (ii) FALSE [Heating carbonate ore is Calcination]
- **Q24**: (i) catenation | (ii) soap
- **Q25**: (i) FALSE [Enters right atrium] | (ii) FALSE [Secreted by pancreas]
- **Q26**: (i) XX | (ii) Fallopian tube (Oviduct)

- **Q27**: (i) $\rightarrow$ (a) Pink colour | (ii) $\rightarrow$ (b) Biomagnification
- **Q28**: (i) Parallel combination. (ii) Total power $= 1000 + 100 + 100 = \mathbf{1200}\ \mathrm{W} = 1.2\ \mathrm{kW}$. (iii) Current $I = \frac{P}{V} = \frac{1200}{220} \approx \mathbf{5.45}\ \mathrm{A}$. (iv) Energy $= 1.2\ \mathrm{kW} \times 5\ \mathrm{h} = \mathbf{6}\ \mathrm{kWh}\ (\text{units})$. (v) Earth wire (Green/Yellow).

Section B Detailed Solutions

Q29:
Newton's Second Law: The rate of change of momentum of a body is directly proportional to the applied unbalanced force and takes place in the direction of the force.
Formula: $F = ma = \frac{m(v-u)}{t}$.

Q30:
1. *Mass*: Quantity of matter in a body; constant everywhere; scalar unit kg.
2. *Weight*: Gravitational pull $W = mg$; varies from place to place; vector unit N.

Q31:
$u = -30\ \mathrm{cm}, f = -15\ \mathrm{cm}$.
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \implies \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{-15} - \frac{1}{-30} = -\frac{1}{15} + \frac{1}{30} = -\frac{1}{30}$.
$\mathbf{v = -30}\ \mathrm{cm}$. The image is real, inverted, and formed at the centre of curvature.

Q32:
1 Dioptre is the power of a lens whose focal length is 1 metre ($1\ \mathrm{D} = 1\ \mathrm{m}^{-1}$).
A power of $-1.5\ \mathrm{D}$ indicates a **Concave Lens** (diverging lens).

Q33:
$\mathrm{Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O}$.
Use: As a disinfectant for water purification or bleaching agent in textile mills.

Q34:
Sodium is an extremely reactive alkali metal. It reacts vigorously with atmospheric oxygen and water vapour, producing hydrogen gas and catching fire. To prevent this, it is stored submerged under kerosene oil.

Q35:
An homologous series is a group of organic compounds sharing the same functional group and general formula, where consecutive members differ by a $-\mathrm{CH_2}-$ unit ($14\ \mathrm{u}$).
First two alkanes: Methane ($\mathrm{CH_4}$) and Ethane ($\mathrm{C_2H_6}$).

Q36:

Neuron diagram consists of:

- Dendrites (receive signals)
- Cyton / Cell body (contains nucleus)
- Axon (transmits nerve impulses)
- Nerve endings (synaptic transmission).

Q37:

10% Law: During energy transfer across trophic levels in an ecosystem, only about 10% of the energy is stored as biomass in the next level; the remaining 90% is dissipated as heat or used in respiration.

Q38:

Glucose ($6\mathrm{C}$) breaks down in cytoplasm into Pyruvate ($3\mathrm{C}$):

1. *In Yeast (Absence of $\mathrm{O_2}$)*: Ethanol + $\mathrm{CO_2}$ + $2\ \mathrm{ATP}$.
2. *In Muscle Cells (Lack of $\mathrm{O_2}$)*: Lactic acid + Energy (causes muscle fatigue).
3. *In Mitochondria (Presence of $\mathrm{O_2}$)*: $\mathrm{CO_2} + \mathrm{H_2O} + 38\ \mathrm{ATP}$.

Q39:

Formula: $\mathrm{CaSO_4 \cdot \frac{1}{2}H_2O}$.

Preparation: $\mathrm{CaSO_4 \cdot 2H_2O} \xrightarrow{373\ \mathrm{K}} \mathrm{CaSO_4 \cdot \frac{1}{2}H_2O} + 1\tfrac{1}{2}\mathrm{H_2O}$.

Uses: 1. Bone fracture casts. 2. Statues and false ceilings.

Q40:

Cross: $TT \times tt \rightarrow F_1 : Tt$ (All Tall).

Selfing: $Tt \times Tt \rightarrow F_2 : TT, Tt, Tt, tt$.

Phenotypic ratio: $3:1$ (3 Tall : 1 Dwarf). Genotypic ratio: $1:2:1$.

Q41:

(a) Electric power is the rate at which electrical energy is consumed in an electric circuit: $P = VI = I^2R$.

(b) Heat $H = I^2Rt = (5)^2 \times 20 \times 30 = 25 \times 600 = \mathbf{15,000}$ Joules ($15\ \mathrm{kJ}$).

Q42:

1. *Roasting*: Heating sulphide ores in excess air:

$2\mathrm{ZnS} + 3\mathrm{O_2} \xrightarrow{\Delta} 2\mathrm{ZnO} + 2\mathrm{SO_2}$.

2. *Calcination*: Heating carbonate ores in limited/no air:

$\mathrm{ZnCO_3} \xrightarrow{\Delta} \mathrm{ZnO} + \mathrm{CO_2}$.

Q43:

(a) Myopia is short-sightedness. Causes: excessive curvature of lens, elongation of eyeball.

(b) Ray diagrams:

- Defective eye: rays converge in front of retina.
- Corrected eye: concave lens diverges rays to focus sharply on retina.

(OR)

(a) $V = IR$. Circuit diagram with battery, key, ammeter, voltmeter, resistor, rheostat.

(b) Connect $3\,\Omega$ and $6\,\Omega$ in parallel: $\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6} = \frac{1}{2} \implies R_p = 2\,\Omega$.

Then connect with $2\,\Omega$ in series: $R_{\text{total}} = 2 + 2 = \mathbf{4\,\Omega}$.

Q44:

(a) Nephron: Bowman's capsule, Glomerulus, Convoluted tubule, Collecting duct.

(b) 1. Ultrafiltration in glomerulus $\rightarrow$ 2. Selective reabsorption of glucose, water, amino acids $\rightarrow$ 3. Tubular secretion of wastes.

(OR)

(a) Catenation (carbon bonds with carbon to form chains/rings) and Tetravalency (valency 4 allows bonding with 4 atoms).

(b) Soap molecule has hydrophilic head ($-\mathrm{COO^-Na^+}$) and hydrophobic tail (hydrocarbon). In water, tails dissolve in central grease droplet while heads face outward, forming a spherical **Micelle** that lifts dirt into water.